Data Augmentation

Deep Learning

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Contents of This Video

In this video, we will cover:

- Why data augmentation helps
- Common augmentation techniques for images
- How augmentation acts as regularization
- Common augmentation parameters
- Best practices and guidelines

The Data Hunger Problem

Deep networks need lots of data:

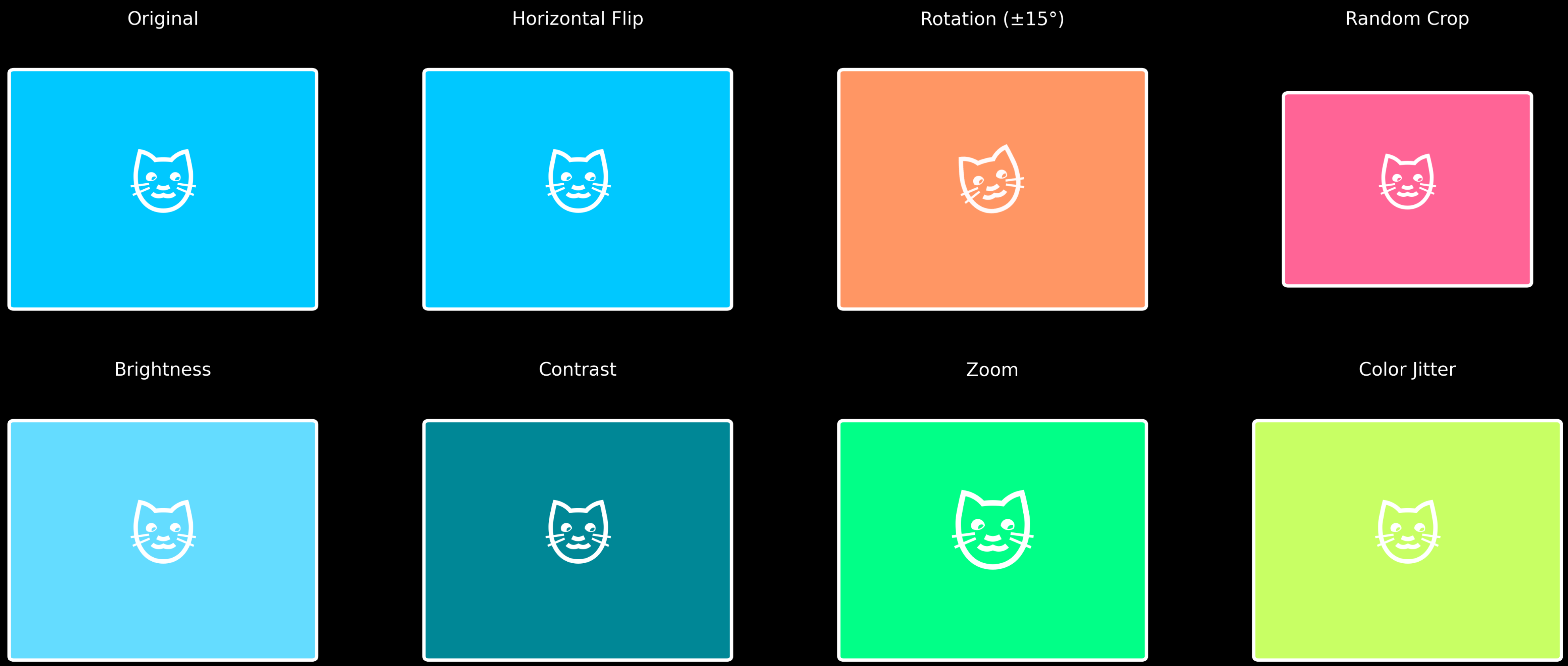
- Small datasets → overfitting
- More data → better generalization
- But: collecting data is expensive!

Solution: Create more training examples artificially

A flipped cat is still a cat!

A rotated digit is still that digit!

Common Image Augmentations



How Augmentation Helps

Without augmentation:

- Model sees same images every epoch
- Can memorize specific pixel patterns
- High risk of overfitting

With augmentation:

- Model sees different versions each epoch
- Must learn robust features
- Acts as regularization

Effective dataset size increases dramatically!

Common Augmentation Parameters

Augmentation	Typical Range	Notes
Rotation	±10° to ±30°	Depends on object orientation
Width/Height shift	10-20%	Simulates object position
Horizontal flip	On/Off	Not for text or asymmetric objects
Zoom	0.9 to 1.1	Simulates camera distance
Brightness	0.8 to 1.2	Simulates lighting conditions

Key principle: Apply augmentation only during training, never on validation/test data

Best Practices

Do:

- Apply augmentation only during training
- Choose task-appropriate transforms
- Start with mild augmentation, increase if overfitting
- Combine multiple augmentations

Don't:

- Augment validation/test data
- Use transforms that change the label (e.g., flip digits 6/9)
- Over-augment (distort images too much)

What We've Covered

In this video, we've learned:

- Data augmentation creates synthetic training examples
- Common augmentations: flip, rotate, crop, brightness, zoom
- Augmentation acts as regularization, reducing overfitting
- Only augment training data, never validation/test
- Choose task-appropriate transformations